

Neurons

Cellular components of the brain's circuitry

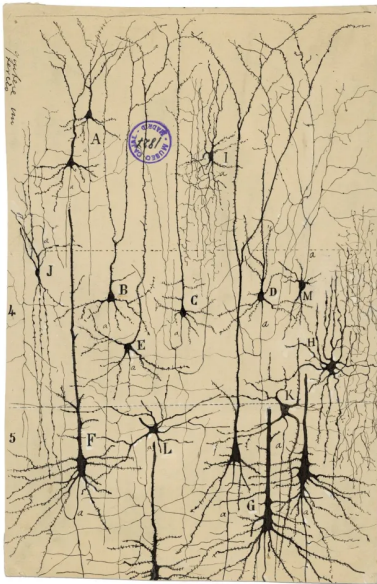
Computational Neuroscience

University of Bristol

M Rule

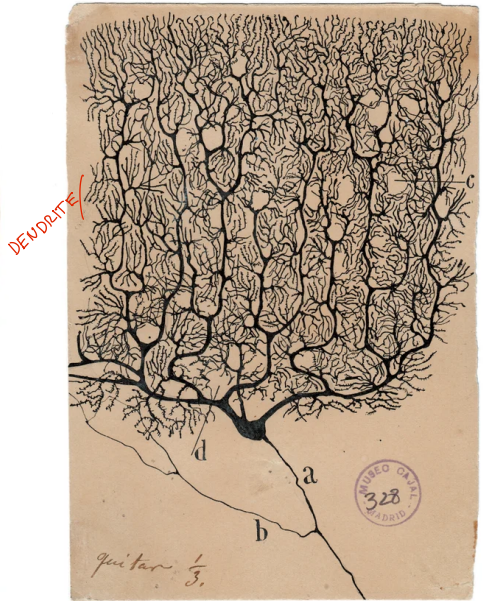
Learning outcomes:

- ▶ Understand the structure of neurons at a basic level
- ▶ Recognize, draw, label soma/axons/dendrites/synapse
- ▶ Describe how neurons communicate with each other
- ▶ Detail the role of Na^+ , K^+ , Cl^- ions in the membrane voltage



Layer V Pyramidal Cells, Santiago Ramón y Cajal

Pyramidal cell soma typically 15–50 μm in diameter



Purkinje Cell, Santiago Ramón y Cajal

Neurons are cells specialised for computation and communication

Soma cell body

- ▶ Nucleus, DNA → mRNA
- ▶ Convergence of electrical signals

Dendrites detect inputs

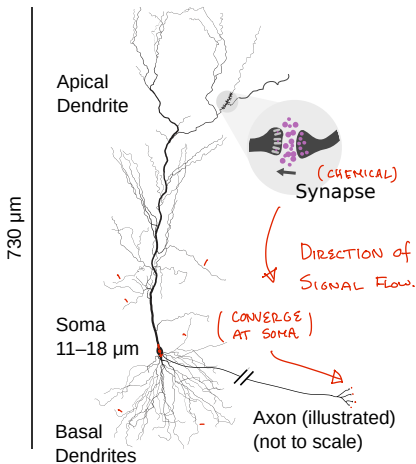
- ▶ Tree-shaped, ~5 – 1000 branches
- ▶ Short-range, local
- ▶ Collect, integrate inputs
- ▶ Analog computation

Axons send output

- ▶ Usually 1 per cell
- ▶ Transmit many targets
- ▶ Local ≤ 1 mm or long-range (~ 1 m in humans)

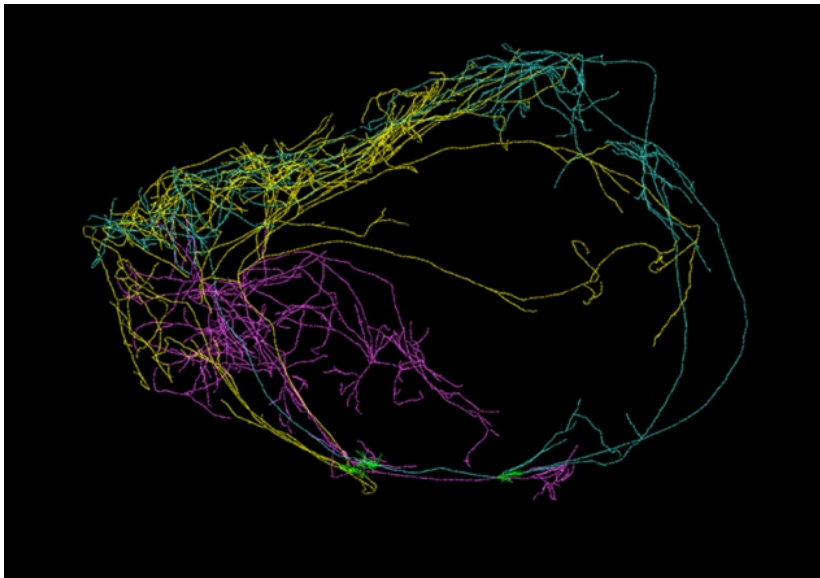
Synapses connect neurons

- ▶ $\sim 10^{14}$ in our brains ($\sim 10^{11}$ neurons)
- ▶ Each neuron has “a few” up to $\sim 10^5$



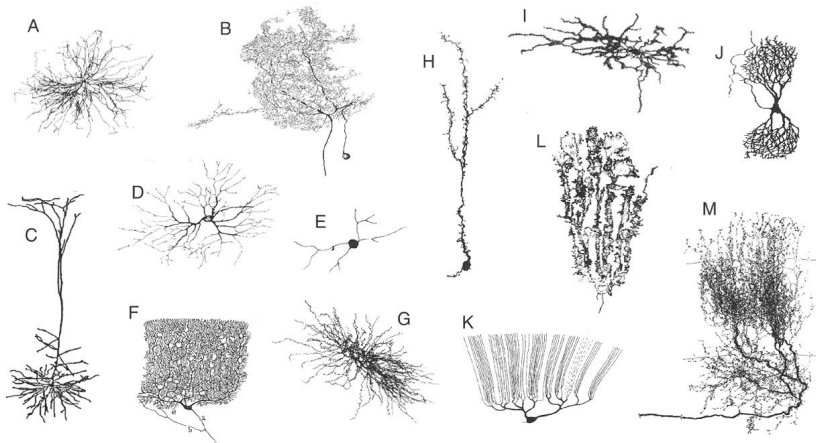
Golding et al. (2005) CA1 Pyramidal Neuron

Axons



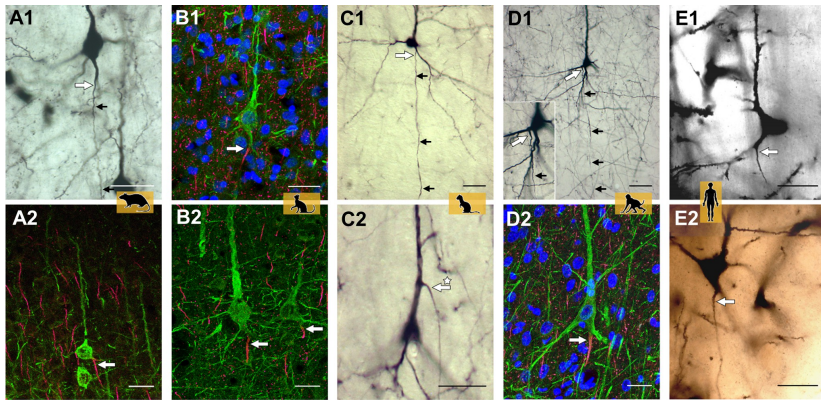
A digital reconstruction of a neuron that encircles the mouse brain.

Dendrites



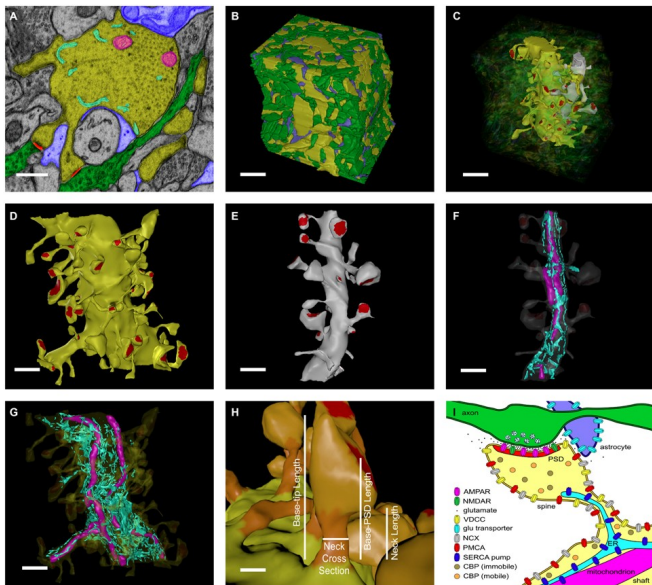
Koch, Biophysics of Computation (1999)

Axons that emerge from dendrites?!



Wahle & al. Neocortical pyramidal neurons with axons emerging from dendrites are frequent in non-primates, but rare in monkey and human (2022)

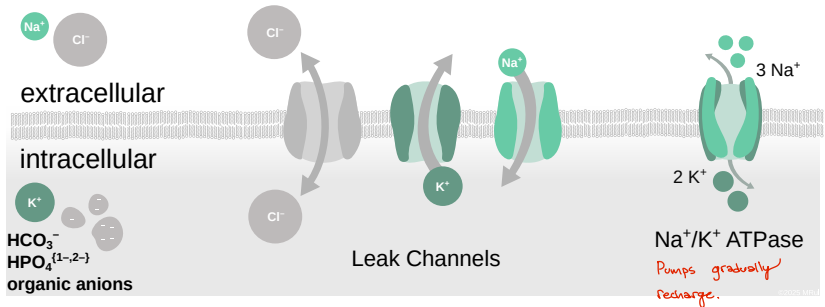
Synapses



Bartol et al(2015) Figure 1. Reconstruction of hippocampal neuropil.

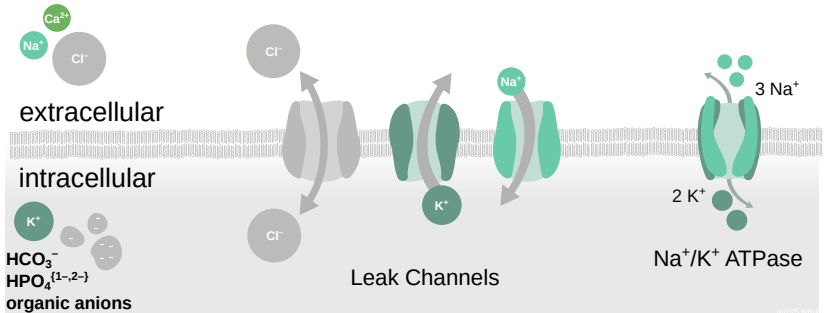
Neurons are excitable

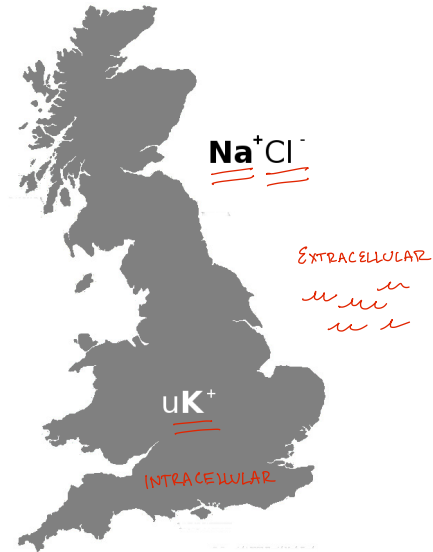
- ▶ They maintain a voltage difference across their cell membrane
- ▶ Charged ions with differing intracellular/extracellular concentrations → chemical “batteries”
 - Effective voltage depends on ion's charge & concentration difference
 - Powers voltage dynamics for computation & communication
 - Active pumps set and maintain concentration differences



Neurons are excitable

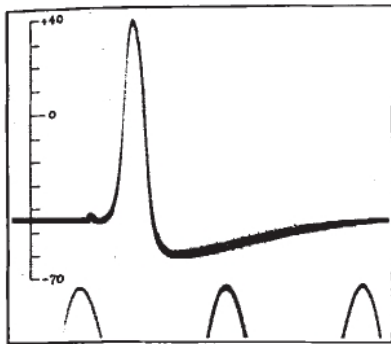
- ▶ They maintain a voltage difference across their cell membrane
- ▶ Charged ions with differing intracellular/extracellular concentrations → chemical “batteries”
 - Effective voltage depends on ion's charge & concentration difference
 - Powers voltage dynamics for computation & communication
 - Active pumps set and maintain concentration differences





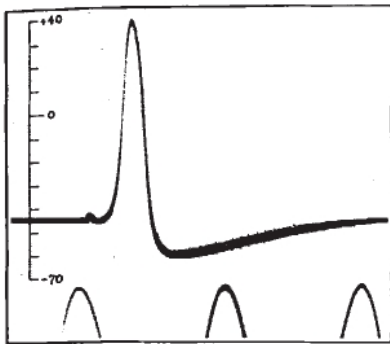
Conor Houghton

Neurons use spikes for reliable long-range communication



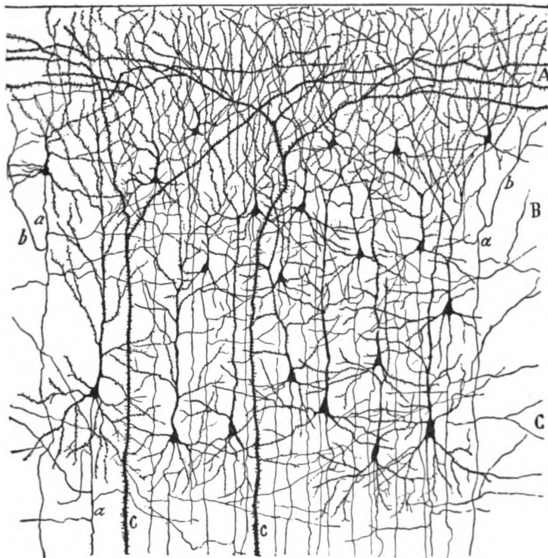
Oscilloscope trace from *Action Potentials Recorded from Inside a Nerve Fibre*, Hodgkin and Huxley (1939) ***“Action potential recorded between inside and outside of axon. Time marker 500 cycles/sec. The vertical scale indicates the potential of the internal electrode in millivolts, the sea water outside being taken at zero potential.”***

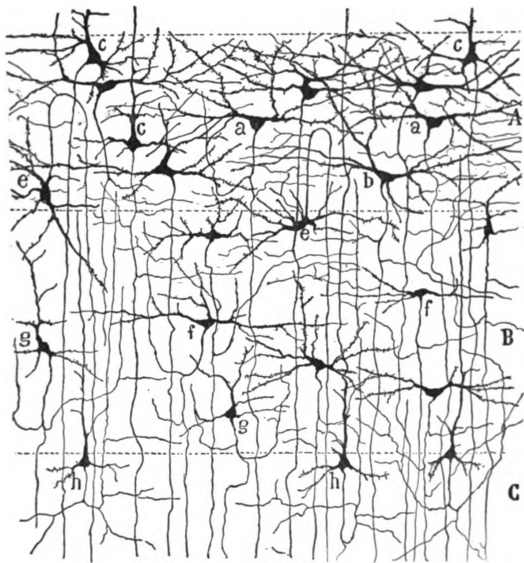
Neurons use spikes for reliable long-range communication

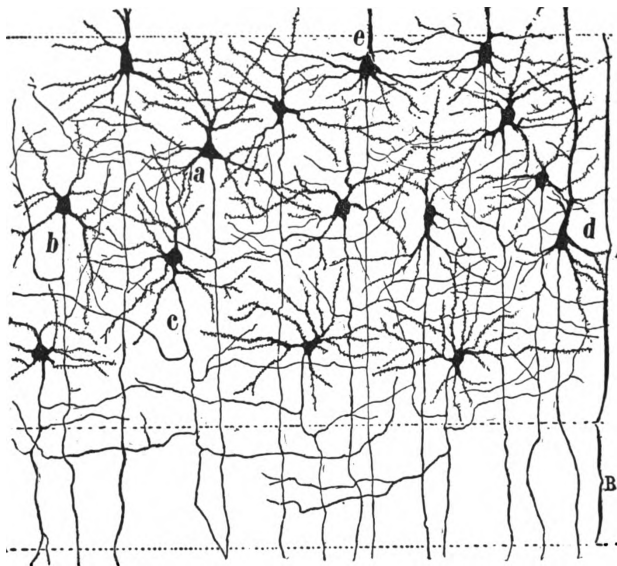


The action potential ("spike") is a sharp rise and fall in cell's membrane voltage
It is a regenerative wave & can travel along axons/dendrites without decay
~ All-or-nothing, binary $\{0, 1\}$

draw and label a prototypical (inter)neuron







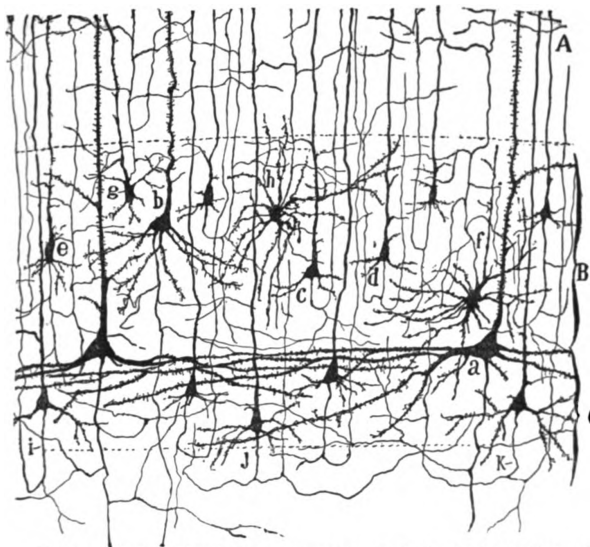
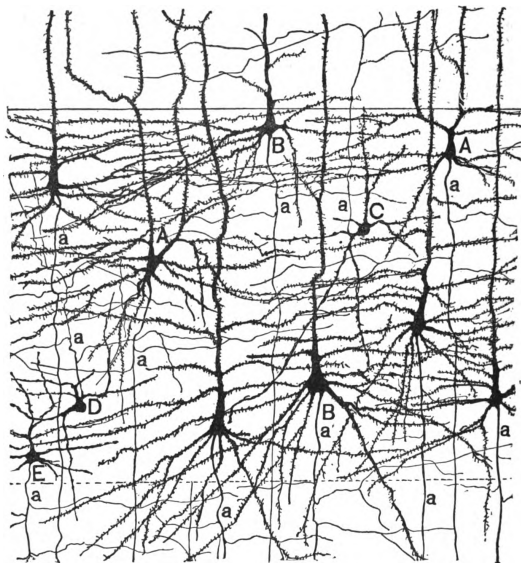


FIG. 20. — Cells of the 6th and 7th layers from the human visual cortex, infant 15 days old.



end

